AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M251-M257
Object Oriented Programming Using Java

Final Examination
Fall Semester 2024/2025

Date: 11 / Jan. / 2025

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Number of Exam Pages: 4 (including this cover sheet)	Time Allowed: 3 hours

nstructions:

1. This exam consists of three parts. It is required to solve **ALL** of them in the external answer booklet:
   - **Part 1: 5** MCQ (10 marks)
   - **Part 2: 3** short answer questions (35 marks)
   - **Part 3: 2** problem solving questions (55 marks)
2. Go through all questions before you start answering them.
3. It is not allowed to use any external material.
4. ıt is not allowed to use any electronic device (calculators, mobiles, …)
5. At the end of the examination:
   - Check that you have written **your name, student ID and section number** on the first page of your answer booklet
   - Place your question paper in the answer booklet and hand over them to the invigilator.

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PART 1: MCQ [10 Marks: 5x2]

Question 1: [10 marks] Read the following questions and choose the best answer.

1. Given the ArrayList `list = [2, 7, 0, 4]`.
   What are the contents of `list` after the following operation? `list.set(2, 4);`
   a. [4, 7, 0, 4] b. [2, 7, 0, 2] c. [2, 7, 4, 4]
   d. [2, 4, 0, 4] e. [2, 7, 4, 0, 4]

2. Which of the following **should not** have abstract methods?
   a. concrete class b. abstract class c. interface
   d. A and B e. B and C

3. Code in the _____ clause will be executed at the end of the try statement, **whether or not** execution was interrupted by an exception.
   a. catch b. finally c. final
   d. A and B e. None of the given choices

4. _____ places the nodes in the top, right, bottom, left and center regions.
   a. Borde-Pane b. GridPane c. FlowPane
   d. HBox e. VBox

5. In Java, the _____ makes decisions about which thread to run next.
   a. follower b. organizer c. distributer
   d. scheduler e. None of the given choices

PART 2: Short Answer Questions [35 Marks]

Question 2: [12 marks: 2+2+2+2+4]

Given the following code, answer the given questions:

```
public abstract class Phone {
  public void hello() {
    System.out.println("Gaza");
  }
  public void price() {
    System.out.println(140);
  }
}
```

```
public class Mobile extends Phone {
  public void price() {
    System.out.println(300);
  }
  public void memory() {
    System.out.println(256);
  }
}
```

```
public class Test {
  public static void main(String[] a
    Phone p = new Mobile();
    Mobile m = new Mobile();
    // valid or not?
    //1// p.memory();
    //2// Phone fone = new Phone();
    p.hello();
    m.hello();
    p.price();
    m.price();
  }
}
```

A. The relationship between the classes `Mobile` and `Phone` is called _____.
B. _____ method is an example of overriding.
C. **Is** the code in the line that starts with the comment `//1//` **valid? Explain.**
D. **Is** the code in the line that starts with the comment `//2//` **valid? Explain.**
E. Provide the exact output of the class `Test`.

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Question 3: [13 marks]

Assume that you have a map called **phones** which includes names of your friends and their phone numbers (as **Strings**), and the phone number length is 8 digits. You noticed that some phone numbers are old, and their length is only 7 digits. It is required to complete the following method to update only the old phone numbers (their length is 7) by adding the digit 2 at the beginning; For example, if the old phone of one of your friends is "6333444", then after calling the method his/her phone will be updated to "26333444".

```
public static ..... oldNumbers(..... phones) {
.....
.....
}
```

Question 4: [10 marks]

Write a **recursive** method that takes as an argument a word and returns how many 'a' characters are in this word; For example, if the word is "banana", the method should return 3.

PART 3: Problem Solving Questions [55 Marks]

Question 5: [42 marks]

1. Write the class **Book** which includes: **[15 marks: 2+3+2+2+4+2]**
   a) 3 attributes: ISBN (e.g. 0-201-50352-X), title (e.g. Algorithms) and author (e.g. Hisham).
   b) a zero-arg constructor and a multi-arg constructor.
   c) getters for **title** and **author** attributes.
   d) **toString()** method which overrides the Object's toString() method.
   e) **equals()** method which overrides the Object's equals() method. It is enough to compare the Book ISBN.
   f) **compareTo()** method which implements the Comparable's compareTo() method. You should use the Book ISBN only in the comparison.

2. Write the class **Library** which includes: (add the necessary **import** statements) **[22 marks: 1+2+5+7+7]**
   a) 2 attributes: Library name and Set of **Books**.
   b) one-arg constructor that accepts the Library name as an argument and creates a new empty Set of **Books** (the set should be sorted automatically according to the Book ISBN).
   c) method **addBook()** that accepts a **Book** object as an argument and adds it to the Set with a confirmation message. If the Book is already in the Set, an error message should be displayed.
   d) method **countBooks()** that takes an author name as an argument and returns how many books in the library are authored by this author.
   e) method **saveIntoFile()** which accepts a file name as an argument. It stores the Library name and information of all Books of the Library in this file (each Book in a different line).

3. Write a code to test your classes by performing the following tasks: **[5 marks: 1+2+1+1]**
   a) create a **Library** object called **myLibrary** where the Library name is "**AOU Library**".
   b) create 2 **Books** and add them to **myLibrary** using the method **addBook()**
   c) print how many books in the Library are authored by "Hisham" by calling the method **countBooks()**
   d) call the method **saveIntoFile()** to save the required information.

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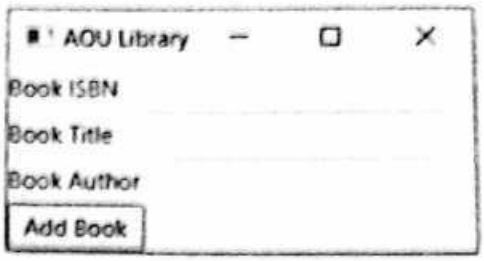
Question 6: [13 marks: 7 + 6]

Based on the classes of Part 3 - Question 5,

Consider the following code which displays the given GUI application and then answer the questions below it:

```
public void start(Stage primaryStage) {
    //Missing Code (1): Initialization
     Library myLibrary = ..........
     Label bookISBNL = ..........
     Label bookTitleL = ..........
     Label bookAuthorL = ..........
     TextField bookISBNTF = ..........
     TextField bookTitleTF = ..........
     TextField bookAuthorTF = ..........
     Button addB = ..........
     GridPane pane = ..........
     primaryStage.setTitle(..........);
     pane.addColumn(0, bookISBNL, ..........., ..........., ..........);
     pane.addColumn(1, bookISBNTF, ..........., ..........);
     Scene scene = new Scene(..........);
     primaryStage.setScene(..........);
     primaryStage.show();

    //Missing Code (2): Clicking on the button
    ..... .....
    ..........
    ..........
```



Complete the first missing code to initialize the UI controls.

Complete the second missing code such that when the user fills in the required data and clicks the button **"Add Book"**, a `Book` object will be created (using this data) and added to `myLibrary`.

End of Questions

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